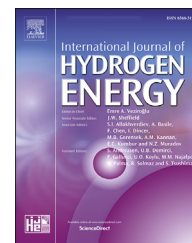


Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/he

Conjoint effect of microwave irradiation and metal nanoparticles on biogas augmentation from anaerobic digestion of green algae

Asad A. Zaidi^a, Feng RuiZhe^a, Adil Malik^a, Sohaib Z. Khan^{b,c},
Asad J. Bhutta^a, Yue Shi^{a,*}, Kashif Mushtaq^d

^a College of Power and Energy Engineering, Harbin Engineering University, Harbin 150001, PR China

^b Department of Mechanical Engineering, Faculty of Engineering, Islamic University of Madinah, Madinah, P. O. Box 170, Saudi Arabia

^c Department of Engineering Sciences, PN Engineering College, National University of Sciences and Technology, Karachi 75350, Pakistan

^d Massachusetts Institute of Technology Portugal Program, Faculty of Engineering, University of Porto (FEUP), Portugal

ARTICLE INFO

Article history:

Received 4 December 2018

Received in revised form

29 January 2019

Accepted 28 February 2019

Available online 16 May 2019

Keywords:

Anaerobic digestion

Biogas

Green algae

Kinetic models

MW pretreatment

Nanoparticles (NPs)

ABSTRACT

Pretreatment of biomass is a commonly applied technique for improving its biodegradability; however, such methods are energy intensive, which affects the overall efficiency. This study aims to provide an energy efficient solution by combining microwave (MW) pretreatment of algal biomass (*Enteromorpha*) and metal nanoparticles (NPs). The MW pretreatment of the biomass was in the form of a slurry (liquid:solid 20:1), while pretreatment time and MW pretreatment power were 6min and 600 W, respectively. Nickel (Ni) and Cobalt (Co) NPs with a concentration of 1 mg/L were used. Batch-wise anaerobic digestion was carried out for a period of 264 h. The results showed that MW pretreatment initiates early hydrolysis of green algae thus reducing lag time. NPs had a positive influence in biogas production at the later stages of anaerobic digestion. The highest total biogas production of 53.60ml/gTS was attained by Co NPs + MW pretreatment whereas maximum biohydrogen of 59.52% (v/v) was produced by Ni NPs + MW pretreatment group. Energy analysis showed that combined utilization of MW pretreatment and metal NPs produced added energy while consuming less input energy than MW pretreatment alone. The kinetic parameters were calculated by using modified Gompertz and Logistic function model for each experimental case.

© 2019 Hydrogen Energy Publications LLC. Published by Elsevier Ltd. All rights reserved.

Introduction

The availability of energy can be enhanced by adopting effective conversion measures of the available resources.

Effective utilization of energy conversion methods supplemented by improved energy utilization techniques has resulted in significant savings in the developed world [1]. Anaerobic digestion (AD) is considered one of the most

* Corresponding author.

E-mail address: shiyue@hrbeu.edu.cn (Y. Shi).

<https://doi.org/10.1016/j.ijhydene.2019.02.245>

0360-3199/© 2019 Hydrogen Energy Publications LLC. Published by Elsevier Ltd. All rights reserved.

dynamic procedures to transform organic matter to a useful form of energy, i.e., biogas. The organic material when confined in a place, in the absence of oxygen, give rise to a large number of bacteria, which assimilate and break down organic matter to produce methane gas as a major by-product [2].

The AD process comprises of four phases. Hydrolysis is the first phase of the AD, during this phase large complex organic polymers present in biomass broken down into simple monomers such as sugar. This stage also termed as the rate-limiting stage of the AD process. During the second phase (acidogenesis), these monomers further transformed into volatile fatty acids (VFAs), ammonia, carbon dioxide (CO₂) and hydrogen sulfide (H₂S) by the action of acidogenic bacteria. The third phase is called acetogenesis, where VFAs under the action of acetogens are converted into acetic acid, hydrogen (H₂) and CO₂. The fourth and terminal phase of the AD is called methanogenesis. During this stage, the output from the preceding stage is acted upon by methanogenic bacteria which transformed intermediate products into methane (CH₄), CO₂ and water.

Algae have been considered as one of the potential feedstock for biogas generation by AD process [3]. However, their cell wall limits the AD process during hydrolysis. The highly resistant and a rigid cell wall restricts anaerobic bacteria to acquire intercellular polymeric components present (carbohydrates, proteins, and cellulose) [4]. Therefore, in order to improve the biodegradability of algal biomass by dissolution of its cell wall and release of polymeric substances, numerous pretreatment techniques have been applied. Kind of algae species and their structure is responsible for different pretreatment efficiencies for the same pretreatment technique employed [5]. However, selection of a pretreatment process is mainly reliant on its low capital cost, positive energy balance and lesser operational cost to make AD process economically feasible [6]. Microwave (MW) pretreatment is the transmission of electromagnetic energy in the frequency range of 0.3–300 GHz. MW pretreatment involves no contact amongst the source and the chemicals [7]. In comparison with the conventional heating methods used, MW pretreatment provides higher heating efficiencies by direct interaction of microwaves with algae cells' wall [8]. Passos et al. [9] studied the effect of MW pretreatment on algae form High Rate Algal Ponds (HRAP). Results showed that MW pretreatment enhanced biogas production rate (25–75%) and successfully improved the digestibility of biomass. In another study conducted by the same authors [10], the continuous reactor was used for the AD of algal biomass from HRAP. In order to improve the biodegradability of algal biomass, MW pretreatment (900 W, 3min) was carried out. Results showed an increase of 30% in methane yield at a hydraulic retention time (HRT) of 15 days and 58% at HRT of 20 days. Several studies discussed MW pretreatment applied to waste activated sludge [11–16]. Almost all the studies reported an enhancement in sludge solubilization and biogas generation.

Enteromorpha is a kind of red tide which severely pollutes the sea. It is rich in protein, carbohydrate, crude fiber and mineral, which can be degraded by micro-organisms anaerobically. Therefore, *Enteromorpha* can be used to produce hydrogen through AD. In our previous study [17], optimization of MW pretreatment for an AD of *Enteromorpha* was carried out

using response surface methodology. Results showed that 24.4 ml biogas/g dry algae were produced at the optimized MW pretreatment conditions after AD.

Minerals are needed for microorganism development [18]. Qiang et al. [19] stated that in the presence of iron (Fe), cobalt (Co), and nickel (Ni), methanogenic bacteria grow fast during AD. Addition of trace metals in the form of NPs improves the efficiency of the biogas production process and the degree of biomass degradation [20]. The effect of Ni and Co NPs with different concentrations (0.5, 1 and 2 mg/L) on biogas and methane yield from the AD of cattle manure slurry was investigated by Abdelsalam et al. [21]. The study indicated that the addition of 1 mg/L Co NPs and 2 mg/L Ni NPs increased the volume of biogas by 1.64 and 1.72 times, respectively. Elreedy et al. [22] also studied the effect of Ni NPs (size of the NPs was 60 nm) at different concentrations (20, 30, 60, and 100 mg/L) on the AD of industrial wastewater containing Mono-Ethylene Glycol (MEG). Results showed that 60 mg/L Ni NPs produce a significant increase of 23% in the hydrogen production.

Our recent study [23] investigated the influence of Ni and Co NPs on the AD of algal biomass (*Enteromorpha*). Results showed that augmentation of NPs in the biomass had a similar effect as that of the control sample during hydrolysis stage; however, a notable incremental effect on the biogas and hydrogen production was observed after 60 h of the digestion process. It has been suggested that combining the pretreatment with NPs may result in an early dissolution of the algae cell wall and provide faster action by NPs on stimulation of microorganisms to achieve high cumulative biogas yield with positive energy balance. Working in the same direction, Khalid et al. [24] applied alkaline pretreatment in combination with magnetite NPs on anaerobic digestion of rice straw. The results showed an increase of 57% in biogas yield. Their findings suggest that the synergism of alkaline pretreatment and magnetite NPs application could effectively enhance biogas yield, and provide a novel approach for optimization of biogas production. Therefore, the objective of the present study is to examine the effect of combining microwave (MW) pretreatment of *Enteromorpha* with Co and Ni NPs. Measurement of soluble indexes was carried out, and the energy ratio was calculated. Established prediction models are also used to substantiate the experimental results of this work for the approximation of the biogas generated during the AD.

Materials and methods

Raw material

Anaerobic sludge was acquired from an Anaerobic-Anoxic-Oxic (AAO) reactor at Harbin Wenchang Sewage Treatment Plant, Harbin, Heilongjiang province, China. Total suspension solids (TSS) of sludge were 6,390 mg/L whereas Volatile Suspension Solids (VSS) were 2,545 mg/L. The *Enteromorpha* was attained from the Institute of Hydrobiology of The Chinese Academy of Science, Wuhan, China. It was air-dried in the drying oven and then sealed in a bottle with a breathable film on the top. The protein, fat and ash content of *Enteromorpha* were 13.20, 1.06 and 21.77%, respectively. Each biodigester contained 60 ml of sludge and 20 g of *Enteromorpha* [17]. Co and Ni NPs (spherical

shaper with an average size < 100 nm) were purchased from China Metallurgical Research Institute, Beijing, China. The concentration of Co and Ni NPs was 1 mg/L. Similar NPs concentration has been used in our previous study [23] and other studies [21,25]. In order to reduce the agglomeration, NPs suspensions were prepared by adding distilled water containing sodium dodecylbenzene sulfonate (SDS) 0.1 mM [26].

Experimental setup

The MW pretreatment was performed before AD with a household Panasonic microwave oven (1180 W). The algae solution was stirred after every minute. The MW pretreatment condition was liquid:solid, pretreatment time and pretreatment power of 20:1, 6min and 600 W, respectively [17]. The laboratory glass bottles (working volume = 500 ml) were used as bioreactors and operated for 264 h. The bioreactors were airtight with rubber plugs. Nitrogen gas was purged through a digester for 5 min at the start to create anaerobic condition [27]. The digesters were retained at 37 °C [25] and 150 rpm mixing speed.

Analytical methods

Total Solids (TS), Volatile Solids (VS), Chemical Oxygen Demand (COD), Soluble Chemical Oxygen Demand (SCOD), Total Chemical Oxygen Demand (TCOD) and reducing sugar were assessed as per the standard methods [28]. 3,5-dinitrosalicylic acid (DNS) method using spectrophotometer (DR 3900, HACH, USA) at a maximum wavelength (λ_{max}) of 550 nm was used to determine glucose concentration. A 50 ml syringe was used to take out biogas samples for measurement of hydrogen, and 0.5 ml biogas was injected into the gas chromatography (SP-2100A, BFL, USA) with Thermal Conductivity Detector-Flame Ionization Detector (TCD-FID) to determine hydrogen content % (v/v) of the biogas. A medical syringe with a long needle was used to collect the samples from air-tight bottles and transferred to small tubes covered with rubber stoppers to avoid gas loss. The biogas generation was measured twice a day whereas its composition was observed once. Each experiment was conducted in triplicate to reduce likely errors, and the average values are indicated. Origin Pro 8 software was used to perform one-way ANOVA analysis of results, $p < 0.05$ was considered to be statistically significant.

Cell lysis rate and energy balance analysis

The cell lysis rate (the extent of cell wall breakage and release of intracellular components) due to the action of NPs and MW pretreatment combined effect during anaerobic digestion was calculated using Eq. (1) [29]:

$$\text{Cell Lysis Rate (\%)} = \frac{\text{SCOD}_T - \text{SCOD}_0}{\text{TCOD}_0 - \text{SCOD}_0} \times 100 \quad (1)$$

Where SCOD_T represented SCOD release during the AD, SCOD_0 and TCOD_0 represented SCOD and TCOD values, respectively, of raw *Enteromorpha*.

The energy assessment was evaluated via calculation of energy input needed for MW pretreatment and the enhancement in hydrogen yield for pretreated *Enteromorpha*. The input

energy and output energy were calculated using Eq. (1) and Eq. (2) and (3) [6]. The energy ratio (Eq. (4)) was calculated as the energy output over energy input. If energy ratio value is greater than 1, it means that the energy yield from hydrogen generation during AD was higher in comparison with the energy required for MW pretreatment. It should be noted that this energy analysis does not include the energy required to dry biomass and other processes for precursors.

$$E_i = \frac{P \times t}{V \times \text{TS}} \quad (2)$$

Where:

E_i = Energy input (kJ/gTS).

P = Power required for MW pretreatment (W).

t = MW pretreatment time (Sec).

V = Volume of biomass (L).

TS = Total solid in biomass (g TS/L).

$$E_o = \frac{\Delta P \times \epsilon}{10^6} \quad (3)$$

Where:

E_o = Energy output (kJ/gTS).

ΔP = Hydrogen yield (ml H_2 /gTS).

ϵ = Calorific value of hydrogen (120,000 kJ/m³).

$$\Delta E = \frac{E_{\text{Hydrogen}}}{E_{\text{Microwave}}} \quad (4)$$

Mathematical kinetic models

The AD process performance with the combined effect of MW pretreatment and metal NPs was mathematically evaluated via modified Gompertz model Eq. (5) [30] and Logistic Function model Eq. (6) [31]. OriginPro 8 software was used to determine kinetic parameters for both models. The software uses an iterative method by employing the Levenberg-Marquardt (L-M) algorithm to estimate parameters for describing reaction kinetics. Akaike Information Criterion (AIC) test was performed to assess which model is better describing the kinetics of the AD process [32]. The model with lower AIC value suggests a better fit and predicting capability. For each model, the AIC value and Akaike's weight value was calculated by using Eq. (7) and Eq. (8) [33]:

$$B = B_p \cdot \exp \left(- \exp \left(\text{MBPR} \cdot \frac{2.7183}{B_p} \cdot (\text{BPDT} - t) + 1 \right) \right) \quad (5)$$

$$B = \frac{B_p}{1 + \exp \left[4 \text{MBPR} \frac{\text{BPDT} - t}{B_p} + 2 \right]} \quad (6)$$

Where:

B = Cumulative biogas volume at digestion time t (ml).

B_p = Biogas production potential (ml).

MBPR = Maximum biogas production rate (ml/h).

BPDT = Biogas production delay time (hrs).

t = Total digestion time (hrs).

$$\text{AIC} = \begin{cases} N \ln \frac{\text{RSS} + 2K}{N}, & \text{when } \frac{N}{K} \geq 40 \\ N \ln \frac{\text{RSS}}{N} + 2K + \frac{2K(K+1)}{N-K-1}, & \text{when } \frac{N}{K} < 40 \end{cases} \quad (7)$$

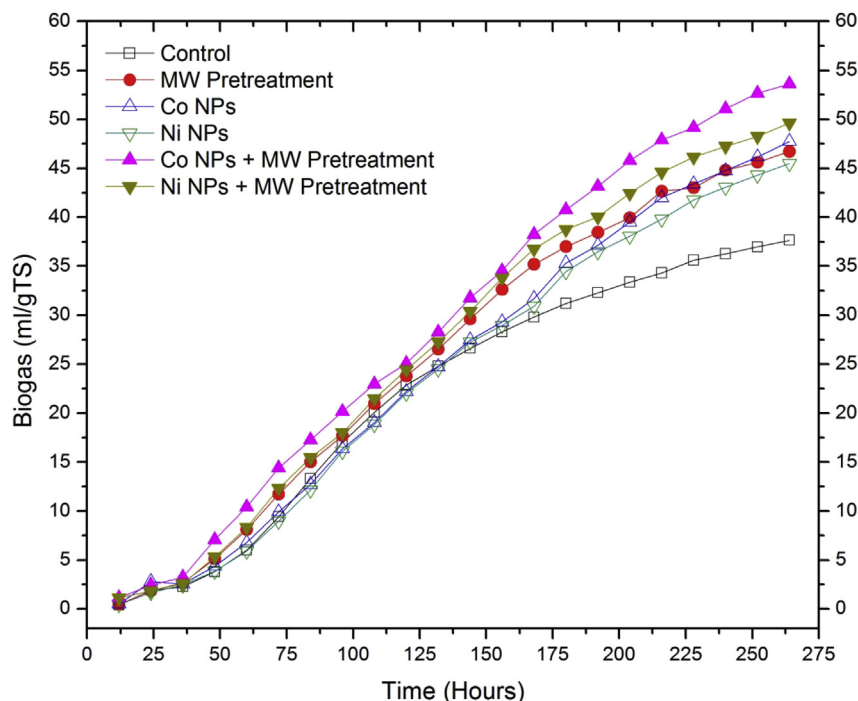


Fig. 1 – Biogas production influenced by different treatments.

$$\text{Akaike's weight} = \frac{e^{(-0.5\Delta\text{AIC})}}{1 + e^{(-0.5\Delta\text{AIC})}} \quad (8)$$

Where:

N = Number of points.

RSS = Residual sum of square.

K = Number of model parameters.

ΔAIC = The relative difference between the two AIC values.

Results and discussion

Effect on biogas and hydrogen production

Fig. 1 illustrates the cumulative biogas production during AD of green algae. It is clear that all treatment methods increase biogas production as compared to the control sample. The maximum total biogas yield of 53.60 ml/gTS was achieved by Co NPs + MW pretreatment group followed by Ni NPs + MW pretreatment. The Co NPs, Ni NPs and MW pretreatment produced an almost similar amount of total biogas. The algae cell wall is composed of an external and internal layer. The external layer is an electron dense polymeric matrix in which glycoprotein and carbohydrates are presented whereas cellulose and hemicellulose exist in the internal layer [34]. During the initial stage, the increase in biogas in combined NPs and MW pretreatment groups is credited to pretreatment method responsible for the dissolution of the external layer by releasing glycoproteins, carbohydrates and cellulose. MW pretreatment rises the lysis rate which results in the increasing effect on biogas production [35]. MW pretreatment hydrolyzes the glycosidic bond present in carbohydrates and polysaccharides which turns into simple sugars. Later, dissolution of the internal layer occurred by the attack of NPs.

The NPs hydrolyze cellulose into oligosaccharides such as cellobiose and cellodextrin [36]. The carbohydrates, proteins, and lipids released by dissolution of the cell structure of algae are then converted into amino acids, simple sugars, peptides and volatile fatty acids [37]. The maximum cumulative biogas and the amount of hydrogen produced during the experiment are presented in Fig. 2 and Fig. 3. Co NPs + MW pretreatment group produced the highest amount of biogas, and Ni NPs + MW pretreatment produced highest hydrogen content (% v/v). The cumulative increase in biogas yield for Co NPs + MW pretreatment, Ni NPs + MW pretreatment and MW pretreatment was 42.36%, 31.73%, and 24%, respectively. In

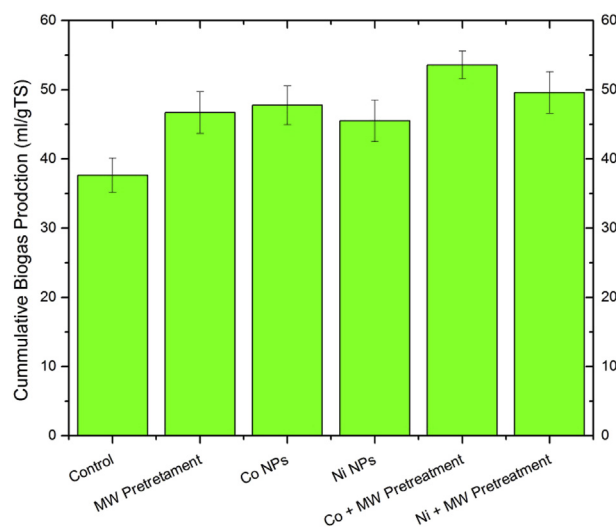


Fig. 2 – Cumulative biogas production (error bars represent standard deviations in observations).

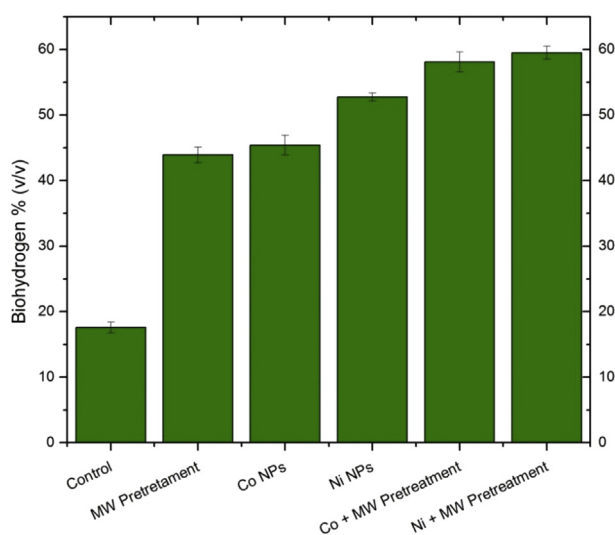


Fig. 3 – Hydrogen content % (v/v) by different treatment conditions (error bars represent standard deviations in observations).

addition, Co NPs + MW pretreatment group resulted in highest the amount of biopolymer release. The carbohydrates and protein release during fermentation is shown in Fig. 4.

To the best of authors' knowledge, this is the first study analyzing the conjoint effect of MW pretreatment and metal NPs for an AD of algae. Biogas production (Fig. 1) suggests that combining MW pretreatment with metal NPs produce a significant effect on biogas enhancement by solubilization of algae cell wall. Our results are in agreement with Passos et al. [9] who reported an increased biogas production rate and a high degree of biomass solubilization by MW pretreatment of algae from HRAP. In another study [38], same authors compared different pretreatment methods for algae anaerobic digestion. Their results showed that MW pretreatment (900 W, 3 min) resulted in

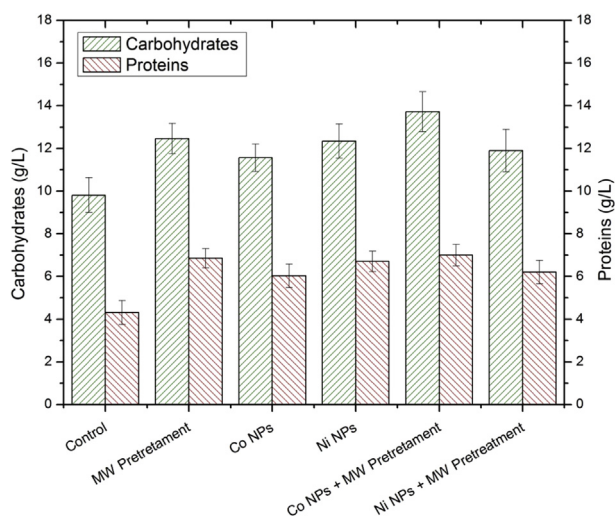


Fig. 4 – Total carbohydrates and proteins produced during fermentation, (error bars represent standard deviations in observations).

an increase of soluble organic matter concentration, soluble protein, and soluble carbohydrates by 8 fold, 18 fold, and 12 fold, respectively. These results are also similar to the results presented in our research, see Fig. 4. Our findings are also in line with Zheng et al. [12], who studied the influence of MW irradiance on primary sludge solubilization. The results showed that MW pretreatment improved soluble chemical oxygen demand (SCOD) in the sludge and the biogas production was enhanced by 37%. Obtained results are also similar with our previous results with *Enteromorpha* [23], in which Ni and Co NPs resulted in 26% and 9% increase in biogas yield, respectively. Abdelsalam et al. [25] also showed an increase in biogas production of 1.8 and 1.7 times by Ni and Co NPs with an AD of cattle manure slurry, respectively. The influence of Ni^{2+} concentration on biogas and hydrogen generation from *Rhodospseudomonas faecalis* RLD-53 was investigated by Liu et al. [39]. The results showed that at a concentration of 236 mg/L Ni^{2+} maximum hydrogen-production of 0.032 g H_2 g/g was produced. The authors mentioned that metal ions stimulate the bacterial activity of hydrogen-producing bacteria inside digester.

Cell lysis and COD concentration

The conjoint effect of MW pretreatment and metal NPs on biodegradability of algal biomass is evaluated by the degree of cell lysis rate (%) shown in Fig. 5. It can be observed that control, Ni NPs and Co NPs had the same degree of cell lysis for the first 60 h of digestion, indicating that NPs were not capable to break down the cell wall. This result is in agreement with our previous study [23]. In contrast, MW pretreatment, Co NPs + MW pretreatment and Ni NPs + MW pretreatment showed early cell lysis. MW pretreatment combined with Co and Ni NPs indicated that more organic matter fraction is present in soluble phase. Co NPs + MW pretreatment group obtained the highest cell lysis of 86.69%.

The conjoint effect of MW pretreatment and metal NPs during AD is evident from the dynamic evolution of COD. It is clear from Fig. 6 that for Co NPs + MW pretreatment, Ni NPs + MW pretreatment and MW pretreatment groups the COD concentration increased for the first 48 h. For rest of the groups, COD increased for 72 h and relatively less steep. COD becomes stable afterwards and starts decreasing at the end. Co NPs + MW pretreatment group provides the highest amount of COD (14,765 mg/L) followed by Ni NPs + MW pretreatment group (13,554 mg/L). The increasing trend by COD values presented algae cell lysis and the release of biopolymeric substances (carbohydrates, glycoproteins, and cellulose) caused by the cell wall dissolution during every treatment. Co NPs + MW pretreatment and Ni NPs + MW pretreatment groups had better COD performance indicating that more oxidizable organic substances are present in these digesters. The rising trend of COD (Fig. 6) for the period of initial 48 h and 72 h is in line with the degree of cell lysis. In this context, it is concluded that MW pretreatment combined with Ni and Co NPs caused an early cell wall disruption and release of biopolymeric segments caused an increase in biogas production. The conjoint effect of Ni and Co NPs with MW pretreatment resulted in increased biogas and biohydrogen yield due to high cell lysis rate, COD and more biopolymers release during AD.

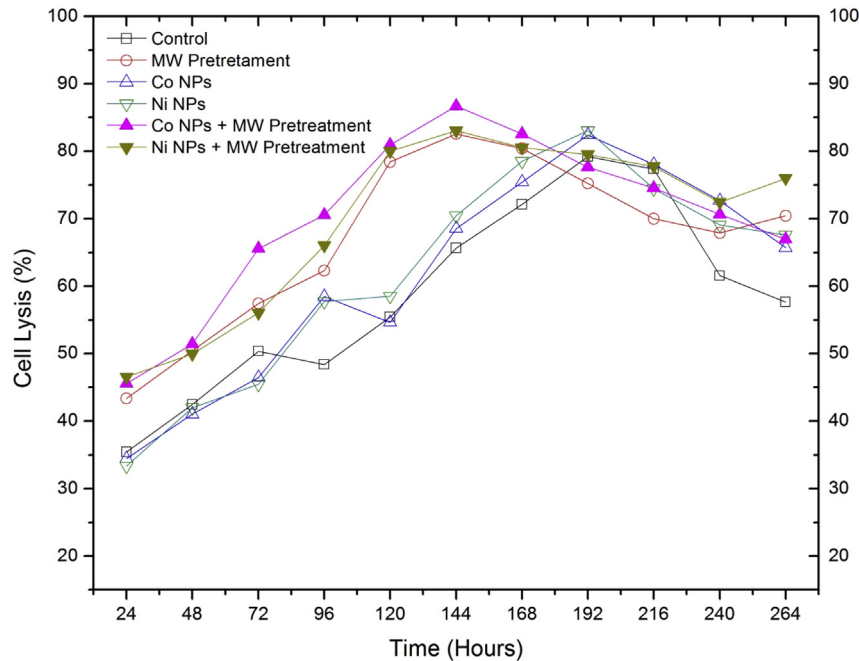


Fig. 5 – Algal biomass cell lysis rate.

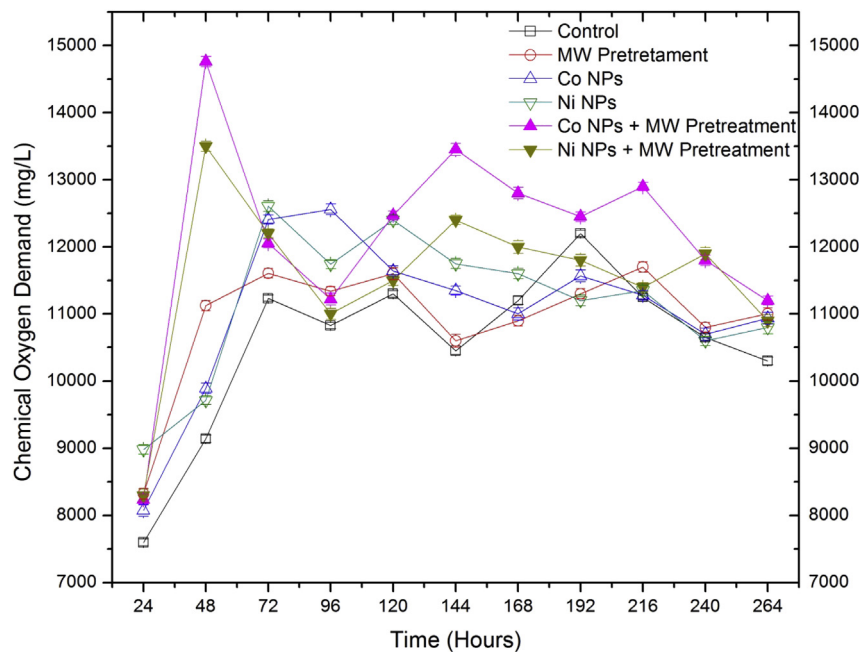
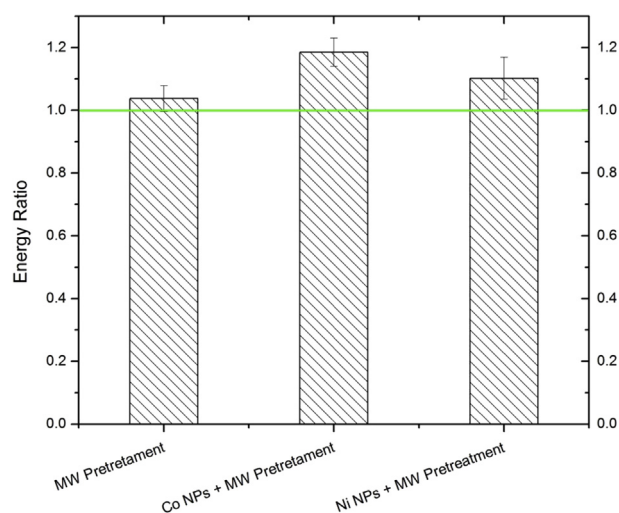


Fig. 6 – Changes in COD concentration during the AD, standard deviations are represented by error bars (error bars represent standard deviations in observations).

Energy assessment

Energy assessment of green algae AD process was conducted for estimating the feasibility of the MW pretreatment, NPs, and their combined effect. For this purpose, the energy generated from algae AD in each case (i.e., output energy, E_o)

was divided by the energy needed for MW pretreatment (i.e., energy input, E_i). The results are shown in Fig. 7 were articulated as the energy ratio (E_o/E_i). It can be observed that for all groups, the energy ratio was higher than one. This indicates that the enhancement in hydrogen production obtained was enough for covering the MW energy input to the AD system,



which may be described by the spontaneity in the AD process after the applied treatments. Co NPs + MW pretreatment group showed the maximum net energy gains by attainment of 1.185 energy ratio.

Kinetic parameters for the cumulative biogas produced by algae AD were found out using modified Gompertz and Logistic Function models [30,31]. The results obtained from the kinetic study using the modified Gompertz and Logistic model are given in Table 1 and Table 2, respectively. Fig. 8 and Fig. 9 showed the comparison of predicted and experimental cumulative biogas yield by all groups. While using the modified Gompertz model, maximum biogas production rate (MBPR) for the non-treated substrate was 0.265ml/gTS.hr. For MW pretreatment, Co NPs, Ni NPs, Co NPs + MW pretreatment and Ni NPs + MW pretreatment, the MBPR found to be 0.276, 0.256, 0.259, 0.282 and 0.284ml/gTS.hr, respectively. Similarly, for the Logistic model, the maximum biogas production rate (MBPR) for the untreated, MW pretreatment, Co NPs, Ni NPs, Co NPs + MW pretreatment and Ni NPs + MW pretreatment were 0.273, 0.292, 0.275, 0.276, 0.300 and 0.302 ml/gTS.hr, respectively. It is determined by both the kinetic models that combined effect of MW pretreatment and NPs had improved the biogas generation rate and reduced the lag phase time with respect to other treatment groups. The decrease in lag phase was observed due to early hydrolysis of algae cell walls at the first stage of AD by MW pretreatment. This resulted in a faster consumption of sugar by anaerobic bacteria in later stages of AD. The correlation coefficient for the modified Gompertz

Parameter	Treatments					
	Control	MW Pretreatment	Co NPs	Ni NPs	Co NPs + MW Pretreatment	Ni NPs + MW Pretreatment
B _P (ml/gTS)	37.94	50.37	55.02	50.34	62.16	54.53
MBPR (ml/gTS.hr)	0.265	0.276	0.256	0.259	0.282	0.284
B PDT (Hrs)	1.47	1.39	1.52	1.61	1.18	1.40
R ²	0.99731	0.99875	0.99823	0.99807	0.99735	0.99872
Predicted Biogas Yield (ml/gTS)	36.63	46.18	47.25	44.86	53.58	49.17
Measured Biogas Yield (ml/gTS)	37.65	46.70	47.75	45.50	53.60	49.60
Difference between measured and predicted biogas yield (%)	2.78	1.12	1.05	1.42	0.037	0.874

Remarks: B_p, Biogas production potential; MBPR, Maximum biogas production rate; BPDT, Biogas production delay time; R², Correlation Coefficient.

Parameter	Treatments					
	Control	MW Pretreatment	Co NPs	Ni NPs	Co NPs + MW Pretreatment	Ni NPs + MW Pretreatment
B _P (ml/gTS)	35.99	46.28	48.65	45.46	55.31	49.61
MBPR (ml/gTS.hr)	0.273	0.292	0.275	0.276	0.300	0.302
B PDT (Hrs)	1.67	1.69	1.89	1.94	1.54	1.73
R ²	0.99005	0.99296	0.99277	0.99156	0.99209	0.99350
Predicted Biogas Yield (ml/gTS)	35.69	45.09	46.22	43.83	52.52	48.04
Measured Biogas Yield (ml/gTS)	37.65	46.70	47.75	45.50	53.60	49.60
Difference between measured and predicted biogas yield (%)	5.49	3.57	3.31	3.81	2.05	3.24

Remarks: B_p, Biogas production potential; MBPR, Maximum biogas production rate; BPDT, Biogas production delay time; R², Correlation Coefficient.

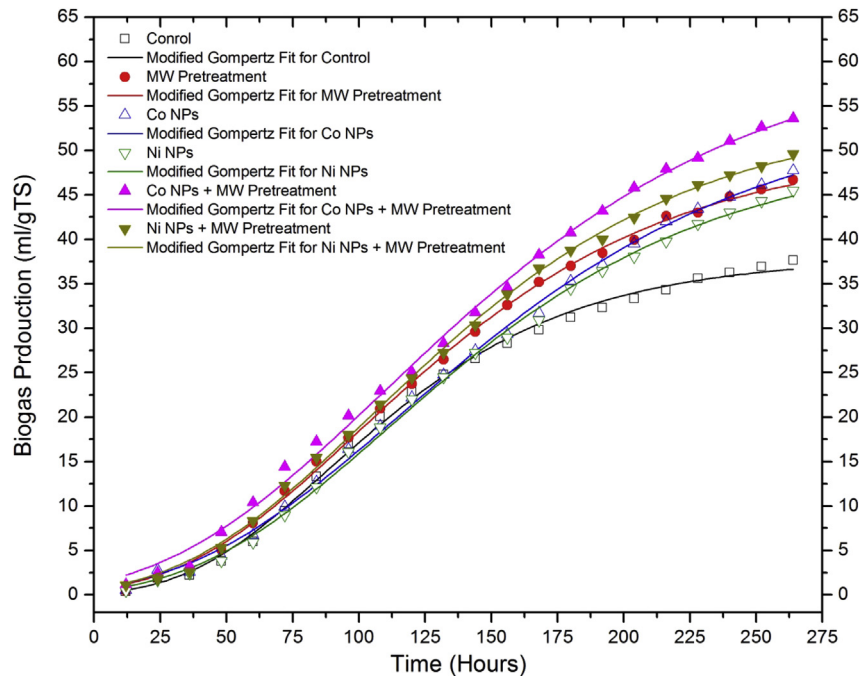


Fig. 8 – Modified Gompertz Fit for experimental data.

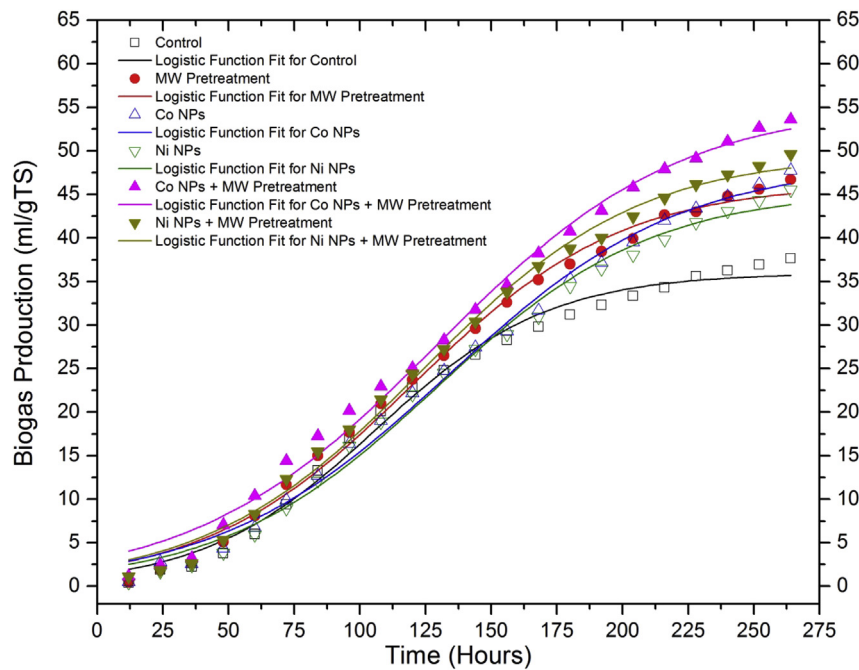


Fig. 9 – Logistic Function Fit for experimental data.

Table 3 – Results for Akaike's information criterion (AIC) test.

Model	RSS	N	AIC	Akaike Weight
Modified Gompertz Model	8.67184	22	-10.12821	1
Logistic Function Model	32.11321	22	18.67389	5.56805E-7
Remarks: RSS, the Residual sum of the square; N, Number of Points; AIC, Akaike's Information Criterion.				

model and Logistic Function model were above 99.73% and 99.00%, respectively. This suggests that both the models were fitting well with the experimental data. Table 3 shows the obtained results for the Akaike Information Criterion (AIC) test. AIC suggests that the modified Gompertz model has a lower AIC value and hence proved to be a better model to use in this case.

Conclusions

The combined effect of MW pretreatment and metal NPs showed the improvement in biodegradability of green algae. The cumulative enhancement in biogas yield for Co NPs + MW pretreatment, Ni NPs + MW pretreatment and MW pretreatment was 42.36%, 31.73%, and 24%, respectively. Highest COD and cell lysis rate of 14,765 mg/L and 86.69% was obtained by Co NPs + MW pretreatment group, respectively. The energy assessment showed the highest energy ratio of 1.185 for Co NPs + MW pretreatment group. The experimental data of these results are further modelled via modified Gompertz and Logistic function model. Akaike Information Criterion (AIC) test highlighted that the modified Gompertz model is nearly matching with the experimental data. This study suggested that positive energy balance occurs when MW pretreatment is combined with metal NPs for an AD of algal biomass.

Acknowledgement

This research was financially supported by the National Key R&D Plan of China (2017YFC1404605), the Natural Science Foundation of China (Grant No. 51579049 and 51509044), the Fundamental Research Funds for the Central Universities (HEUCFG201820), the High Tech Ship Program (17KJ-003) and the Research on 130,000GT cruise ship development.

REFERENCES

- [1] Faiz Ahmed S, Mushtaq K, Ali A. Design and performance analysis of floating dome type portable biogas plant for domestic use in Pakistan-manufacturing cost optimization. *Biotechnology* 2016;15:112–8. <https://doi.org/10.3923/biotech.2016.112.118>.
- [2] Mushtaq K, Zaidi AA, Askari SJ. Design and performance analysis of floating dome type portable biogas plant for domestic use in Pakistan. *Sustain Energy Technol Assessments* 2016;14:21–5. <https://doi.org/10.1016/j.seta.2016.01.001>.
- [3] Salama E-S, Kurade MB, Abou-Shanab RAI, El-Dalatony MM, Yang I-S, Min B, et al. Recent progress in microalgal biomass production coupled with wastewater treatment for biofuel generation. *Renew Sustain Energy Rev* 2017;79:1189–211. <https://doi.org/10.1016/j.rser.2017.05.091>.
- [4] Montingelli ME, Tedesco S, Olabi AG. Biogas production from algal biomass: a review. *Renew Sustain Energy Rev* 2015;43:961–72. <https://doi.org/10.1016/j.rser.2014.11.052>.
- [5] Kendir E, Ugurlu A. A comprehensive review on pretreatment of microalgae for biogas production. *Int J Energy Res* 2018;37:11–31. <https://doi.org/10.1002/er.4100>.
- [6] Córdova O, Passos F, Chamy R. Physical pretreatment methods for improving microalgae anaerobic biodegradability. *Appl Biochem Biotechnol* 2018;185:114–26. <https://doi.org/10.1007/s12010-017-2646-6>.
- [7] Appels L, Houtmeyers S, Degève J, Van Impe J, Dewil R. Influence of microwave pre-treatment on sludge solubilization and pilot scale semi-continuous anaerobic digestion. *Bioresour Technol* 2013;128:598–603. <https://doi.org/10.1016/j.biortech.2012.11.007>.
- [8] Zaidi AA, Feng R, Malik A, Khan ZS, Shi Y, Bhutta JA, et al. Combining microwave pretreatment with iron oxide nanoparticles enhanced biogas and hydrogen yield from green algae. *Process* 2019;7. <https://doi.org/10.3390/pr7010024>.
- [9] Passos F, Solé M, García J, Ferrer I. Biogas production from microalgae grown in wastewater: effect of microwave pretreatment. *Appl Energy* 2013;108:168–75. <https://doi.org/10.1016/j.apenergy.2013.02.042>.
- [10] Passos F, Hernández-Marín M, García J, Ferrer I. Long-term anaerobic digestion of microalgae grown in HRAP for wastewater treatment. Effect of microwave pretreatment. *Water Res* 2014;49:351–9. <https://doi.org/10.1016/j.watres.2013.10.013>.
- [11] Climent M, Ferrer I, Baeza M del M, Artola A, Vázquez F, Font X. Effects of thermal and mechanical pretreatments of secondary sludge on biogas production under thermophilic conditions. *Chem Eng J* 2007;133:335–42. <https://doi.org/10.1016/j.cej.2007.02.020>.
- [12] Zheng J, Kennedy KJ, Eskicioglu C. Effect of low temperature microwave pretreatment on characteristics and mesophilic digestion of primary sludge. *Environ Technol* 2009;30:319–27. <https://doi.org/10.1080/09593330902732002>.
- [13] Solyom K, Mato RB, Pérez-Elvira SI, Cocero MJ. The influence of the energy absorbed from microwave pretreatment on biogas production from secondary wastewater sludge. *Bioresour Technol* 2011;102:10849–54. <https://doi.org/10.1016/j.biortech.2011.09.052>.
- [14] Eskicioglu C, Kennedy KJ, Droste RL. Characterization of soluble organic matter of waste activated sludge before and after thermal pretreatment. *Water Res* 2006;40:3725–36. <https://doi.org/10.1016/j.watres.2006.08.017>.
- [15] Park W-J, Ahn J-H, Hwang S, Lee C-K. Effect of output power, target temperature, and solid concentration on the solubilization of waste activated sludge using microwave irradiation. *Bioresour Technol* 2010;101:S13–6. <https://doi.org/10.1016/j.biortech.2009.02.062>.
- [16] Eskicioglu C, Kennedy KJ, Droste RL. Enhanced disinfection and methane production from sewage sludge by microwave irradiation. *Desalination* 2009;248:279–85. <https://doi.org/10.1016/j.desal.2008.05.066>.
- [17] Feng R, Zaidi AA, Zhang K, Shi Y. Optimisation of microwave pretreatment for biogas enhancement through anaerobic digestion of microalgal biomass. *Period Polytech - Chem Eng* 2018;1–8. <https://doi.org/10.3311/PPCh.12334>.
- [18] Ganzoury MA, Allam NK. Impact of nanotechnology on biogas production: a mini-review. *Renew Sustain Energy Rev* 2015;50:1392–404. <https://doi.org/10.1016/j.rser.2015.05.073>.
- [19] Qiang H, Niu Q, Chi Y, Li Y. Trace metals requirements for continuous thermophilic methane fermentation of high-solid food waste. *Chem Eng J* 2013;222:330–6. <https://doi.org/10.1016/j.cej.2013.02.076>.
- [20] Keskin T, Arslan K, Nalakth Abubackar H, Vural C, Eroglu D, Karaalp D, et al. Determining the effect of trace elements on biohydrogen production from fruit and vegetable wastes. *Int J Hydrogen Energy* 2018;43:10666–77.
- [21] Abdelsalam E, Samer M, Attia YA, Abdel-Hadi MA, Hassan HE, Badr Y. Effects of Co and Ni nanoparticles on biogas and methane production from anaerobic digestion of slurry. *Energy Convers Manag* 2017;141:108–19. <https://doi.org/10.1016/j.enconman.2016.05.051>.
- [22] Elreedy A, Ibrahim E, Hassan N, El-Dissouky A, Fujii M, Yoshimura C, et al. Nickel-graphene nanocomposite as a novel supplement for enhancement of biohydrogen production from industrial wastewater containing mono-ethylene glycol. *Energy Convers Manag* 2017;140:133–44. <https://doi.org/10.1016/j.enconman.2017.02.080>.

- [23] Zaidi AA, RuiZhe F, Shi Y, Khan SZ, Mushtaq K. Nanoparticles augmentation on biogas yield from microalgal biomass anaerobic digestion. *Int J Hydrogen Energy* 2018;43:14202–13. <https://doi.org/10.1016/j.ijhydene.2018.05.132>.
- [24] Khalid MJ, Zeshan, Waqas A, Nawaz I. Synergistic effect of alkaline pretreatment and magnetite nanoparticle application on biogas production from rice straw. *Bioresour Technol* 2019;275:288–96. <https://doi.org/10.1016/j.biortech.2018.12.051>.
- [25] Abdelsalam E, Samer M, Attia YA, Abdel-hadi MA, Hassan HE, Badr Y. Comparison of nanoparticles effects on biogas and methane production from anaerobic digestion of cattle dung slurry. *Renew Energy* 2016;87:592–8. <https://doi.org/10.1016/j.renene.2015.10.053>.
- [26] Khan SZ, Yuan Y, Abdolvand A, Schmidt M, Crouse P, Li L, et al. Generation and characterization of NiO nanoparticles by continuous wave fiber laser ablation in liquid. *J Nanoparticle Res* 2009;11:1421–7. <https://doi.org/10.1007/s11051-008-9530-9>.
- [27] Xia A, Jacob A, Tabassum MR, Herrmann C, Murphy JD. Production of hydrogen, ethanol and volatile fatty acids through co-fermentation of macro- and micro-algae. *Bioresour Technol* 2016;205:118–25. <https://doi.org/10.1016/j.biortech.2016.01.025>.
- [28] *Standard methods for the examination of water and waste water*. 4th ed. Beijing, China: Environmental Press; 2002.
- [29] Kavitha S, Preethi J, Rajesh Banu J, Yeom IT. Low temperature thermochemical mediated energy and economically efficient biological disintegration of sludge: simulation and prediction studies for anaerobic biodegradation. *Chem Eng J* 2017;317:481–92. <https://doi.org/10.1016/j.cej.2017.02.092>.
- [30] Syaichurrozi I, Budiyo Sumardiono S. Predicting kinetic model of biogas production and biodegradability organic materials: biogas production from vinasse at variation of COD/N ratio. *Bioresour Technol* 2013;149:390–7. <https://doi.org/10.1016/j.biortech.2013.09.088>.
- [31] Deepanraj B, Sivasubramanian V, Jayaraj S. Effect of substrate pretreatment on biogas production through anaerobic digestion of food waste. *Int J Hydrogen Energy* 2017;42:26522–8. <https://doi.org/10.1016/j.ijhydene.2017.06.178>.
- [32] George Casella, et al. *Applied regression analysis: a research tool*. Second. Springer-Verlag New York, Inc.; 1998.
- [33] Wong CS, Li WK. A note on the corrected Akaike information criterion for threshold autoregressive models. *J Time Ser Anal* 1998;19:113–24. <https://doi.org/10.1111/1467-9892.00080>.
- [34] Demuez M, Mahdy A, Tomás-Pejó E, González-Fernández C, Ballesteros M. Enzymatic cell disruption of microalgae biomass in biorefinery processes. *Biotechnol Bioeng* 2015;112:1955–66. <https://doi.org/10.1002/bit.25644>.
- [35] Mahdy A, Mendez L, Ballesteros M, González-Fernández C. Enhanced methane production of *Chlorella vulgaris* and *Chlamydomonas reinhardtii* by hydrolytic enzymes addition. *Energy Convers Manag* 2014;85:551–7. <https://doi.org/10.1016/j.enconman.2014.04.097>.
- [36] Ehimen EA, Holm-Nielsen J-B, Poulsen M, Boelsmand JE. Influence of different pre-treatment routes on the anaerobic digestion of a filamentous algae. *Renew Energy* 2013;50:476–80. <https://doi.org/10.1016/j.renene.2012.06.064>.
- [37] Kavitha S, Yukesh Kannah R, Yeom IT, Do K-U, Banu JR. Combined thermo-chemo-sonic disintegration of waste activated sludge for biogas production. *Bioresour Technol* 2015;197:383–92. <https://doi.org/10.1016/j.biortech.2015.08.131>.
- [38] Passos F, Carretero J, Ferrer I. Comparing pretreatment methods for improving microalgae anaerobic digestion: thermal, hydrothermal, microwave and ultrasound. *Chem Eng J* 2015;279:667–72. <https://doi.org/10.1016/j.cej.2015.05.065>.
- [39] Liu B-F, Ren N-Q, Ding J, Xie G-J, Guo W-Q. The effect of Ni²⁺, Fe²⁺ and Mg²⁺ concentration on photo-hydrogen production by *Rhodospseudomonas faecalis* RLD-53. *Int J Hydrogen Energy* 2009;34:721–6. <https://doi.org/10.1016/j.ijhydene.2008.11.033>.